Source-Grounded Multimodal Evidence Review for Public Defenders

Claim-First Pipeline Doubles Discrepancy Detection with Bodycam Video

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NVIDIA Nemotron Omni

Evaluated on L40S GPU

github.com/harishvs/counterpoint

1. Problem & Motivation

Public defenders in the U.S. handle **80%** of criminal cases but are chronically under-resourced. A single attorney may manage hundreds of cases, each with hours of bodycam footage and multi-page incident reports.

The Gap

- Officers' reports may contain discrepancies with video evidence
- Manual cross-referencing takes hours per case
- Missed discrepancies can affect case outcomes

Our Goal

An AI tool that **automatically flags potential conflicts** between incident reports and bodycam footage, with source references for attorney verification.

Non-Goals

This tool does not determine guilt, assess misconduct, decide credibility, or replace attorney review. All findings require human verification.

2. Benchmark Dataset

15	220	9
Scenarios	Claims	Discrepancy Types

Each scenario pairs a **real OCDA bodycam video** with a fictional incident report containing planted discrepancies.

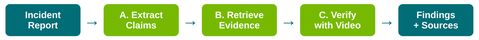
Ground Truth Distribution

Difficulty Levels

Data source: Orange County District Attorney OIS investigation reports. Evidence not redistributed; manifest provides public download URLs with SHA-256 hashes.

3. Claim-First Pipeline Architecture

The pipeline decomposes evidence review into three targeted stages, each operating on focused inputs rather than processing entire videos at once.



A. Claim Extraction

LLM decomposes the report into atomic factual claims (spoken statements, action descriptions, object descriptions, temporal sequences).

B. Evidence Retrieval

Keyword search matches each claim to timestamped transcript segments. **No LLM call** — fast, deterministic, saves API budget for verification.

C. Verification with Video Clips

For each matched claim, ffmpeg cuts a short video clip (under Nemotron's 2-minute limit). The model compares the claim against **actual video + audio evidence**.

Why Claim-First?

- **Nemotron Omni's 2-minute video limit** makes single-pass review of full bodycam videos impossible (typically 5–60+ minutes)
- Targeted clips focus the model on relevant evidence
- Each finding traces to a specific report passage + video timestamp
- Modular stages enable independent improvement

4. Deployment Architecture

Designed for office-controlled AI: DGX Spark is our preferred on-premises deployment target for reviewing sensitive discovery without sending case content to external AI services.

Experimental vs. Proposed Deployment

	Experimental Platform	Proposed Deployment
Hardware	L40S (48 GB VRAM)	DGX Spark (128 GB unified)
Location	Cloud GPU instance	Defender office network
Status	**Evaluated**	Proposed target

Component	Technology
Model	Nemotron 3 Nano Omni 30B-A3B (NVFP4, 21 GB)
Serving	vLLM 0.20 with video+audio support
Input	Video (mp4) + audio + text, up to 2 min per clip
Pipeline	Python, keyword retrieval + ffmpeg clip extraction
License	NVIDIA Open Model Agreement (commercial OK)

Privacy-motivated deployment: the intended design keeps inference and evidence storage local, with access controls and data-handling safeguards subject to office approval. Confidentiality and compliance also depend on system configuration, access controls, and approved operating procedures.

5. Results

Baseline Comparison

Baseline	P	R	F1
Claude Haiku transcript-only	40.0%	2.8%	5.3%
Nemotron Omni **video + transcript**	**62.5%**	**7.1%**	**12.7%**

+56%	+154%	+140%
Precision gain	Recall gain	F1 gain

By Difficulty

Difficulty	P	R	F1	n
Easy	45.5%	10.6%	17.2%	116
Medium	100%	7.5%	13.9%	67
Hard	0%	0%	0%	37

By Discrepancy Type

Type	P	R	n
Fabrication	**100%**	**18.2%**	**11**
Omission	100%	9.1%	11
Minimization	100%	7.7%	13
Direct contradiction	100%	6.0%	83
Exaggeration	0%	0%	12
Ambiguous language	0%	0%	13

6. Confusion Matrix

Nemotron Omni (video + transcript), 220 claims

	ambiguous	not_assessed	conflict	supported
Conflict (141)	**88**	32	**10**	3
Supported (69)	43	12	**6**	**2**
Ambiguous (10)	6	3	0	0

Rows = ground truth, columns = predicted. The model is conservative: it defaults to *ambiguous* rather than guessing, yielding high precision but low recall.

7. Key Findings

- **Video evidence doubles F1** compared to transcript-only review
- **100% precision** on fabrication, omission, minimization, and direct contradiction when flagged
- **Claim-first architecture is necessary**: Nemotron's 2-min video cap prevents single-pass review of full bodycam footage
- **Keyword retrieval outperforms LLM retrieval** for timestamp localization (faster, more accurate)
- 21% abstention rate shows the model appropriately declines when evidence is insufficient
- **DGX Spark as proposed deployment**: NVFP4 model (21 GB) fits within Spark's 128 GB unified memory, enabling on-premises inference where sensitive discovery materials stay within the defender office

8. Limitations & Next Steps

Current Limitations

- Low recall (7.1%) — most claims lack matching transcript keywords for clip retrieval
- Hard discrepancies (0% recall) — subtle timing/sequence errors undetected
- Visual-only claims (positions, objects) poorly served by audio transcript matching
- Benchmark uses authored reports, not real police reports

Next Steps

- Frame-based retrieval for visual claims (periodic keyframe sampling)
- Larger models (FP8/BF16) for improved reasoning
- Validate on DGX Spark and benchmark against L40S results
- Seek defender office or legal-clinic collaborator for task review
- Expand benchmark: American Fork PD, synthetic scenarios, negative examples

References

NVIDIA Nemotron 3 Nano Omni — build.nvidia.com
vLLM — github.com/vllm-project/vllm
OCDA OIS Reports — ocdistrictattorney.gov
Counterpoint benchmark — github.com/harishvs/counterpoint